



Marangoni Flows

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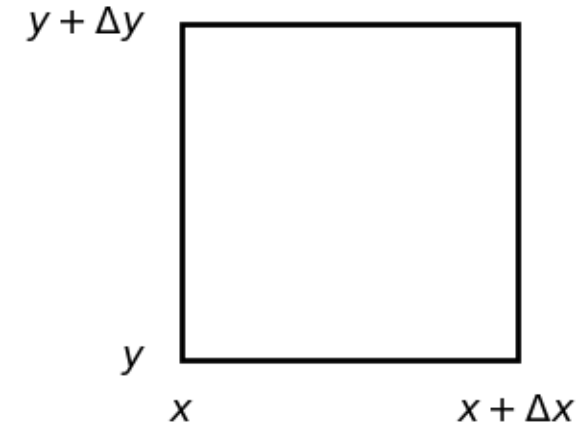
Derivation

- $\mathbf{F} = (\gamma(x + \Delta x)\Delta y - \gamma(x)\Delta y ,$
 $\gamma(y + \Delta y)\Delta x - \gamma(y)\Delta x)$
 $= (\frac{\partial \gamma}{\partial x} \Delta x \Delta y , \frac{\partial \gamma}{\partial y} \Delta x \Delta y)$

$$\text{Stress } \tau = \left(\frac{\partial \gamma}{\partial x}, \frac{\partial \gamma}{\partial y} \right) = \mu \left(\frac{\partial u_x}{\partial z}, \frac{\partial u_y}{\partial z} \right)$$

Characteristic speed u^* :¹

$$u^* = \frac{h}{\mu} \nabla \gamma$$



Classical papers



(1855) Thomson first identified the tears of wine phenomenon as a result of the lower alcohol concentrations above. ¹

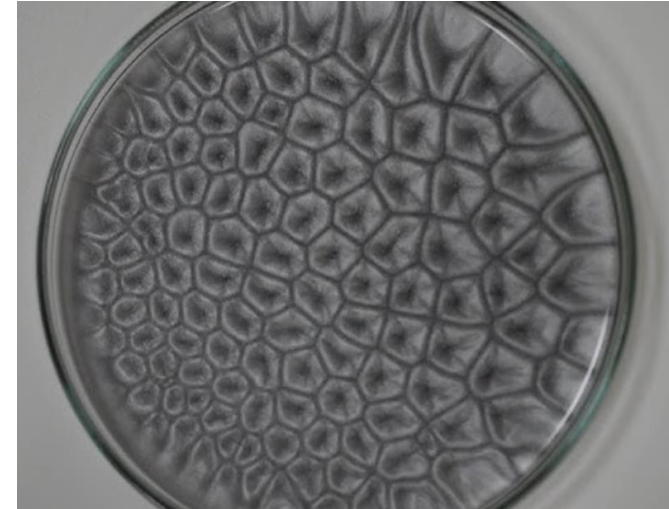
(1865) Marangoni showed that any surface tension gradient would pull liquid from low surface tension to high surface tension²

1. *The London, Edinburgh and Dublin Philosophical Magazine and Journal of Science*. 1855. Taylor & Francis.

2. <https://docnum.unistra.fr/digital/collection/coll12/id/93397/>.

Image: Puiu, Tibi. 2014. "Beautiful Flow Visualization Explains Surface Tension." *ZME Science*, October 23. <https://www.zmescience.com/science/physics/marangoni-flow-surface-tension/>.

Classical papers



(1958) Pearson proved that hexagonal Benard-convection-cells would spontaneously appear when the Marangoni (Marangoni forces/viscous forces) number is large enough.¹

1. Pearson, J. R. A. 1958. "On Convection Cells Induced by Surface Tension." *Journal of Fluid Mechanics* 4 (5): 489–500. <https://doi.org/10.1017/S0022112058000616>.

Image: Fluid Dynamics, dir. 2016. *Benard-Marangoni Instability*. 04:10. <https://www.youtube.com/watch?v=v2vMXmuC818>.

The coffee-ring effect

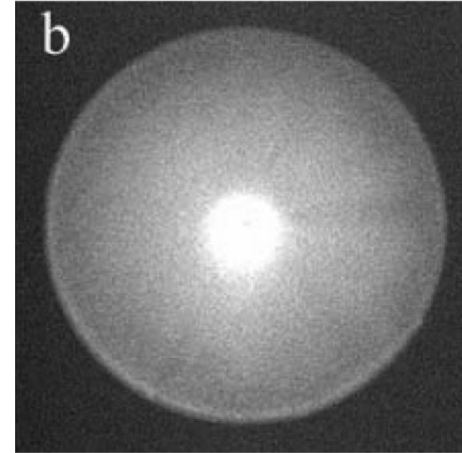


(1997) The coffee-ring effect was first explained by Deegan et. Al as a result of capillary action. Organic solvents assemble at the contact line.¹

1. Deegan, Robert D., Olga Bakajin, Todd F. Dupont, Greb Huber, Sidney R. Nagel, and Thomas A. Witten. 1997. "Capillary Flow as the Cause of Ring Stains from Dried Liquid Drops." *Nature* 389 (6653): 827–29.
<https://doi.org/10.1038/39827>.

Image: *Wikipedia*. 2025. "Coffee ring effect." November 7. https://en.wikipedia.org/w/index.php?title=Coffee_ring_effect&oldid=1320975725.

The coffee-ring effect



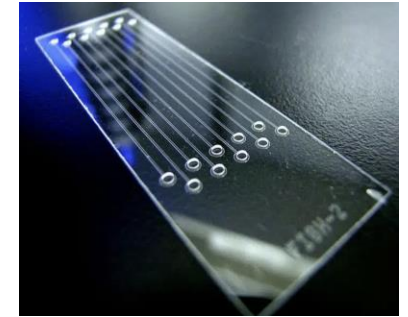
(2006) Hu and Larson showed that the marangoni effect in an interface clean of surfactants could counter the capillary action.¹

Evaporation at the edges is the largest, the latent heat causes a temperature gradient. This gives rise to a Marangoni flow to the center of the droplet.

1. Hu, Hua, and Ronald G. Larson. 2006. "Marangoni Effect Reverses Coffee-Ring Depositions." *The Journal of Physical Chemistry B* 110 (14): 7090–94. <https://doi.org/10.1021/jp0609232>.

Applications

Microfluidchips: Precise local heating can induce accurate flow of microfluids within chips.



Desalination process: works against the accumulation of salt in desalination installations.

Microswimmers: The controlled release of surfactants can create directional motion of active swimmers

Wafer drying for semiconductors: Alcohol creates surface tension gradient to remove water.

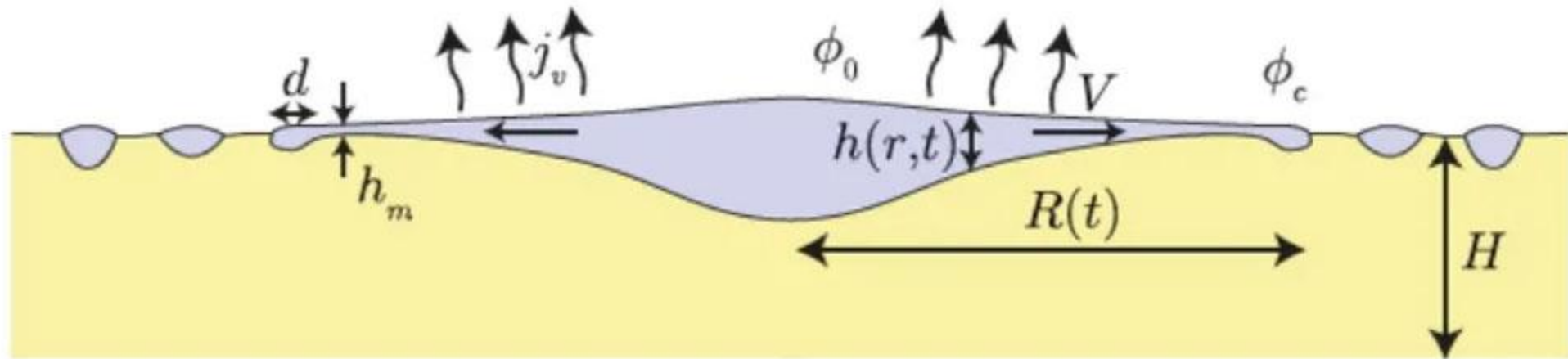


Current research/problems

Accurate modeling of marangoni flows is essential for the development of new/better nanofluid applications.

Interesting research directions are: Machine learning algorithms & Non-Newtonian fluids.

Marangoni Bursting



Alcohol evaporates quicker at the side
→ Surface tension gradient: $\gamma_{rim} > \gamma_0$
→ outward flow

Marangoni Bursting

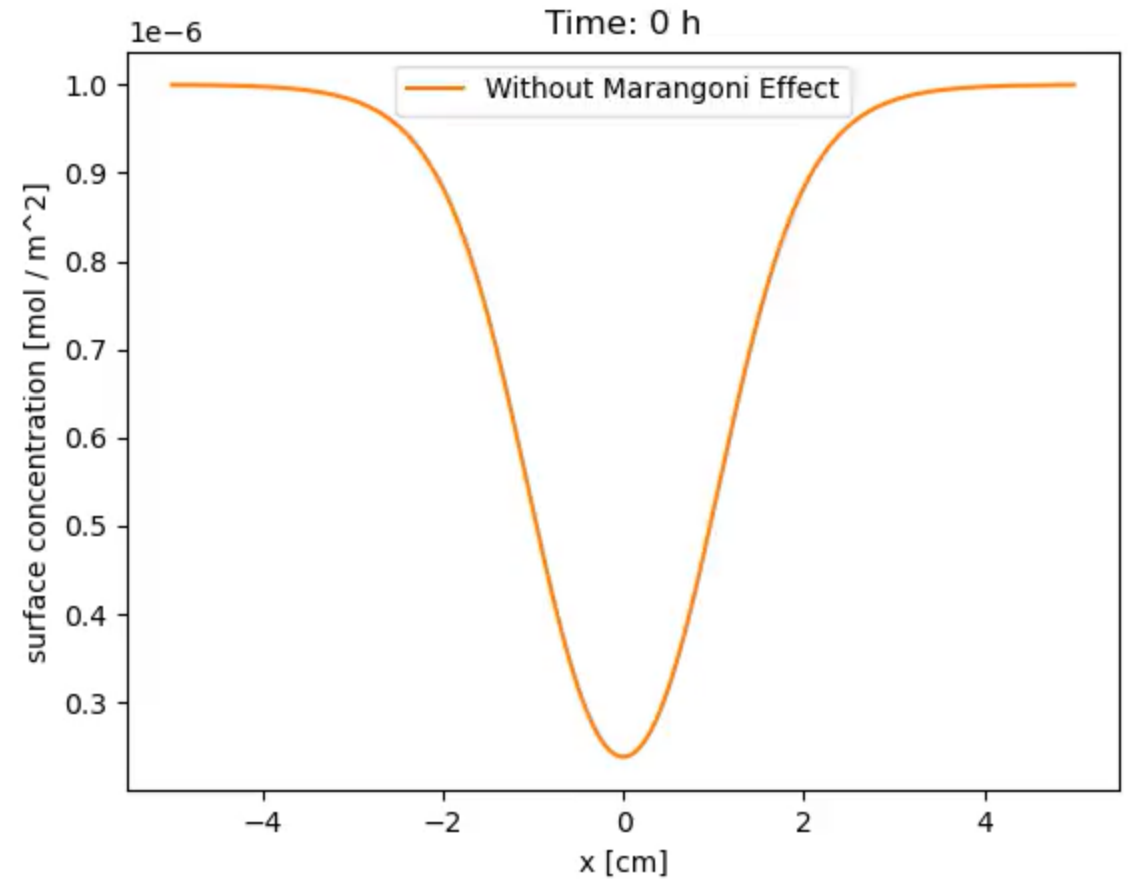
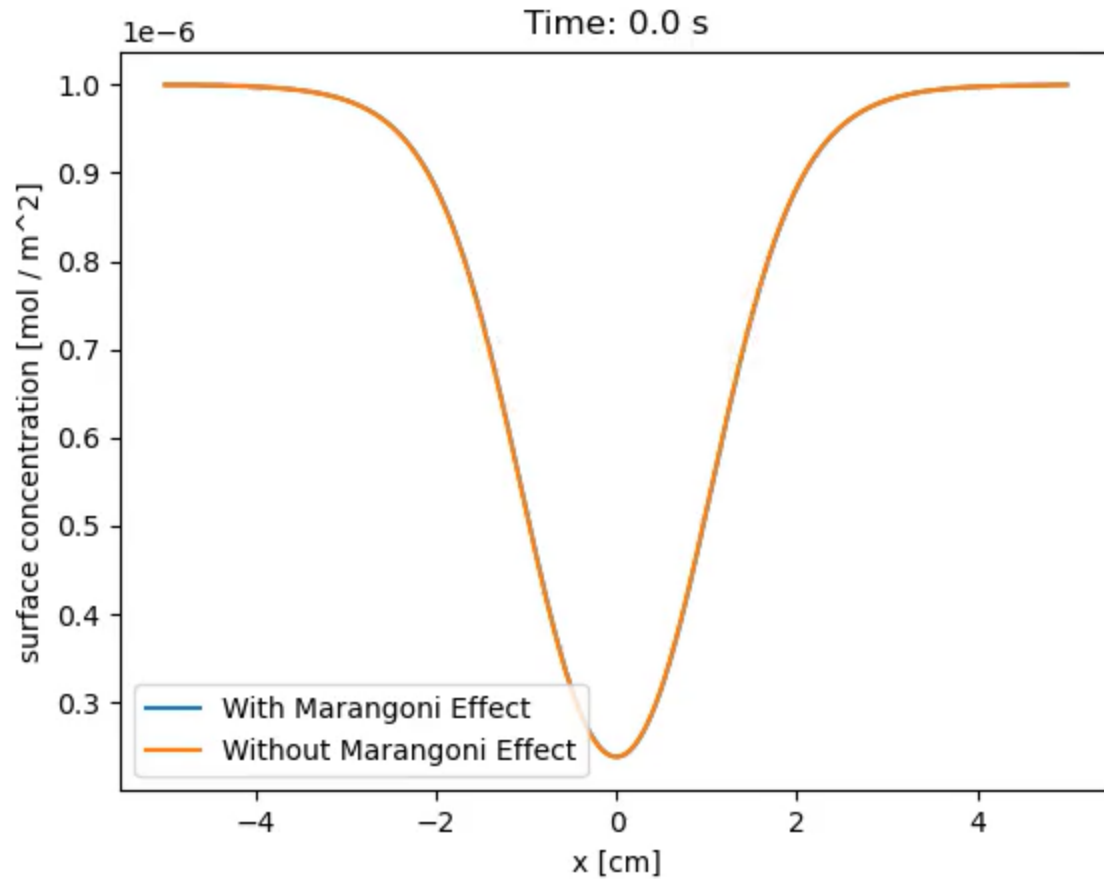
Alcohol-Water Droplet in Oil



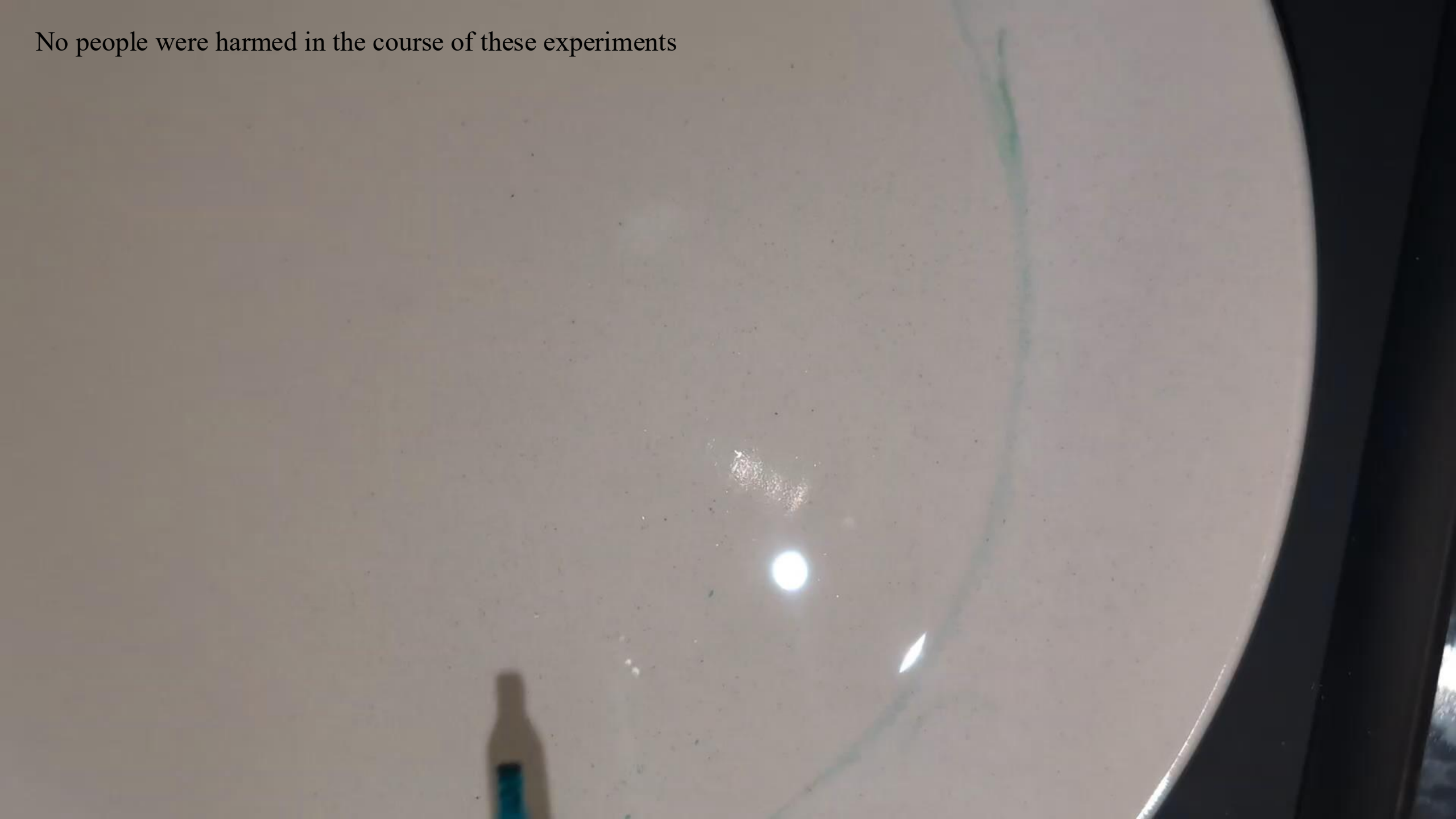
Self-healing Soap Bubbles



Self-healing Soap Bubbles



No people were harmed in the course of these experiments



Calculation Parameters

- Model continuity equation for surface concentration of surfactant Γ :¹
 - $\frac{\partial \Gamma}{\partial t} + \frac{\partial}{\partial x} (J_{diff} + J_{ma}) = 0$ with $J_{diff} = -D \frac{\partial \Gamma}{\partial x}$ and $J_{ma} = \Gamma \frac{h}{\mu} \frac{\partial \gamma(\Gamma)}{\partial x}$
- Henry's Law: $\gamma = \gamma_0 - k_B T \Gamma$ ²
- Specific values:
 - $T = 20^\circ C$, $h = 1 \text{ mm}$, $\Gamma_0 = 1 \times 10^{-6} \frac{\text{mol}}{\text{m}^2}$ ³
 - $\gamma_0 = 72,8 \frac{\text{mN}}{\text{m}}$ ⁴, $\mu = 1 \text{ mPa} \cdot \text{s}$ ⁵, $D = 1 \times 10^{-9} \frac{\text{m}^2}{\text{s}}$ ¹
- Solved with Methods of Lines and RK45

1. Matar, Omar K., and Chris J. Lawrence. 2006. "The Effect of Surfactant on the Flow of a Thin Liquid Film over a Spinning Disc." *Chemical Engineering Science* 61 (4): 1074–91. <https://doi.org/10.1016/j.ces.2005.08.015>.

2. Manikantan, Harishankar, and Todd M. Squires. 2020. "Surfactant Dynamics: Hidden Variables Controlling Fluid Flows." *Journal of Fluid Mechanics* 892 (June): P1. <https://doi.org/10.1017/jfm.2020.170>.

3. Zdziennicka, Anna, Joanna Krawczyk, Katarzyna Szymczyk, and Bronisław Jańczuk. 2018. "Macroscopic and Microscopic Properties of Some Surfactants and Biosurfactants." *International Journal of Molecular Sciences* 19 (7): 1934. <https://doi.org/10.3390/ijms19071934>.

4. Wikipedia. 2025. "Surface tension." November 12. https://en.wikipedia.org/w/index.php?title=Surface_tension&oldid=1321735780.

5. Wikipedia. 2025. "Surface tension." November 12. https://en.wikipedia.org/w/index.php?title=Surface_tension&oldid=1321735780.